

2.15(c) The result here is very useful to construct a random variable given a cdf.

- i. For any $0 < x < 1$, $F(t) \geq x \Rightarrow t \in \{y \in \mathbb{R} : F(y) \geq x\} \Rightarrow t \geq \inf\{y \in \mathbb{R} : F(y) \geq x\} \Rightarrow t \geq F_1^{-1}(x)$. On the other hand, if $t \geq F_1^{-1}(x)$, since there exists a sequence $y_n, n = 1, \dots$, in $\{y \in \mathbb{R} : F(y) \geq x\}$ such that $y_n \geq F_1^{-1}(x)$ and $\lim_{n \rightarrow \infty} y_n = F_1^{-1}(x)$, by right-continuity of CDF F , $F(t) \geq F(F_1^{-1}(x)) = \lim_{n \rightarrow \infty} F(y_n) \geq x$. Thus $F(t) \geq x \Leftrightarrow t \geq F_1^{-1}(x)$. So $P(Z_1 \leq t) = m(\{x : F_1^{-1}(x) \leq t\}) = m(\{x : x \leq F(t)\}) = F(t)$.

Similarly, one can show that $F(t) > x \Rightarrow F_2^{-1}(x) \leq t$, so $F(t) = m\{x : F(t) > x\} \leq m\{x : F_2^{-1}(x) \leq t\}$, and $F_2^{-1}(x) \leq t \Rightarrow F(t) \geq x$, thus $m\{x : F_2^{-1}(x) \leq t\} \leq m\{x : F(t) \geq x\} = F(t)$. Therefore the distribution of Z_2 is still $F(t) \geq x \Leftrightarrow t \geq F_1^{-1}(x)$.

- ii. $\forall \epsilon > 0, \exists \delta_\epsilon > 0$, such that $F(F_1^{-1}(x) - \epsilon) < x - \delta_\epsilon$, because, by definition of F_1^{-1} , $F(F_1^{-1}(x) - \epsilon) < x$. Use the result proved in i., $F(t) \geq x \Leftrightarrow t \geq F_1^{-1}(x)$, we get $F_1^{-1}(x - \delta_\epsilon) > F_1^{-1}(x) - \epsilon$. Thus we proved, $\forall \epsilon > 0, \exists \delta_\epsilon > 0$, such that $x' \in (x - \delta_\epsilon, x) \Rightarrow F_1^{-1}(x) - F_1^{-1}(x') \leq F_1^{-1}(x) - F_1^{-1}(x - \delta_\epsilon) < \epsilon$, i.e. F_1^{-1} is left continuous.

Similarly, $\forall \epsilon > 0, \exists \delta_\epsilon > 0$ such that $F(F_2^{-1}(x) + \epsilon) > x + \delta_\epsilon \Rightarrow F_2^{-1}(x + \delta_\epsilon) \leq F_2^{-1}(x) + \epsilon$. Thus, F_2^{-1} is right continuous.

- 2.20 Let $f_i(x) = \sum_{j=1}^{\infty} a_{ij} I_{[j, j+1)}(x)$, $i = 1, \dots, n$, then f_i is nonnegative measurable function for each i . By Corollary 2.3.5, $\sum_{i=1}^{\infty} \int_{\mathbb{R}} f_i dm = \int_{\mathbb{R}} \sum_{i=1}^{\infty} f_i dm$. $\int_{\mathbb{R}} f_i dm = \sum_{j=1}^{\infty} a_{ij} m([j, j+1)) = \sum_{j=1}^{\infty} a_{ij}$ by using Corollary 2.3.5 again. On the other hand, $\sum_{i=1}^{\infty} f_i(x) = \sum_{j=1}^{\infty} \{(\sum_{i=1}^{\infty} a_{ij}) I_{[j, j+1)}(x)\}$, we won't be bothered if $\sum_{i=1}^{\infty} a_{ij} = +\infty$ for some j , because, in this case, the equality we need to prove holds automatically. So, by Corollary 2.3.5 once again, $\int_{\mathbb{R}} \sum_{i=1}^{\infty} f_i dm = \int_{\mathbb{R}} \sum_{j=1}^{\infty} \{(\sum_{i=1}^{\infty} a_{ij}) I_{[j, j+1)}(x)\} dm = \sum_{j=1}^{\infty} \int_{\mathbb{R}} \{(\sum_{i=1}^{\infty} a_{ij}) I_{[j, j+1)}(x)\} dm = \sum_{j=1}^{\infty} (\sum_{i=1}^{\infty} a_{ij})$. In all, we proved $\sum_{i=1}^{\infty} (\sum_{j=1}^{\infty} a_{ij}) = \sum_{j=1}^{\infty} (\sum_{i=1}^{\infty} a_{ij})$ for nonnegative a_{ij} .

- 2.25 For any finite partition $P = \{x_1, \dots, x_n\}$, since $\mathbb{Q}$ is dense in $\mathbb{R}$, $M_i = \sup_{x_i \leq x \leq x_{i+1}} f(x) = 1$, and $m_i = \inf_{x_i \leq x \leq x_{i+1}} f(x) = 0$ due to that the irrational number is also dense in $\mathbb{R}$. Thus $U(P, f) = 1$, $L(P, f) = 0$, which means that the Riemann integral of f doesn't exist. But the Lebesgue integral of f is $\int_{\mathbb{R}} f(x) dm = m(\mathbb{Q} \cap [0, 1]) = 0$ (countable additivity).

- A. i. The notation $[a] \equiv \max\{n \in \mathbb{Z} : n \leq a\}$, $a \wedge b$ means $\min(a, b)$. Because $f(x) \geq 0$,

$$f_n(x) = \frac{[2^n f(x) \wedge 2^n]}{2^n} = \sum_{i=1}^{n2^n} \frac{i-1}{2^n} \cdot I \left\{ x : x \in f^{-1} \left(\left[\frac{i-1}{2^n}, \frac{i}{2^n} \right) \right) \right\} + n \cdot I \{ x : x \in f^{-1}([n, \infty)) \}$$

By **Corollary 2.1.4** and the fact that $I\{x : x \in f^{-1}(A)\}$ is $(\mathcal{F}, \mathcal{B}(\mathbb{R}))$ -measurable for any $A \in \mathcal{B}(\mathbb{R})$, $f_n(x)$ is a simple nonnegative $(\mathcal{F}, \mathcal{B}(\mathbb{R}))$ -measurable function.

ii. $\forall x \in \Omega, f(x) \geq 0$, then

$$\begin{aligned}
f_{n+1}(x) &= \frac{\lfloor 2^{n+1}f(x) \wedge 2^{n+1}(n+1) \rfloor}{2^{n+1}} = \frac{\lfloor 2 \cdot (2^n f(x) \wedge 2^n(n+1)) \rfloor}{2^{n+1}} \\
&\geq \frac{\lfloor 2 \cdot \lfloor 2^n f(x) \wedge 2^n(n+1) \rfloor \rfloor}{2^{n+1}} = \frac{2 \cdot \lfloor 2^n f(x) \wedge 2^n(n+1) \rfloor}{2^{n+1}} \\
&= \frac{\lfloor 2^n f(x) \wedge 2^n(n+1) \rfloor}{2^n} \\
&\geq \frac{\lfloor 2^n f(x) \wedge 2^n n \rfloor}{2^n} \\
&= f_n(x)
\end{aligned}$$

In the above argument, We only use the simple facts that $\lfloor a \rfloor + 1 > a \geq \lfloor a \rfloor$ and $a > b \Rightarrow \lfloor a \rfloor \geq \lfloor b \rfloor$, which also implies that for any integer x , $\lfloor x \rfloor = x$.

iii. For each fixed x , $\exists n_0 \in \mathbb{N}$ such that $f(x) \leq n_0$, then for all $n \geq n_0$, $f(x) \wedge n = f(x)$, thus

$$|f(x) - f_n(x)| = \left| \frac{2^n f(x) - \lfloor 2^n f(x) \wedge 2^n n \rfloor}{2^n} \right| = \frac{|2^n f(x) - \lfloor 2^n f(x) \rfloor|}{2^n} \leq \frac{1}{2^n}.$$

So $\lim_{n \rightarrow \infty} f_n(x) = f(x), \forall x \in \Omega$.

B. As $\min(f, g) \leq |f| + |g|$, $\min(f, g) \in L_1(\Omega, \mathcal{F}, \mu)$.

Use the expression $\min(a, b) = (a + b - |a - b|)/2$,

$$\begin{aligned}
2 \left\{ \min \left(\int f d\mu, \int g d\mu \right) - \int \min(f, g) d\mu \right\} &= \int (f + g) d\mu - \left| \int (f - g) d\mu \right| - \int (f + g) d\mu + \int |f - g| d\mu \\
&= \int |f - g| d\mu - \left| \int (f - g) d\mu \right| \geq 0.
\end{aligned}$$

With equality holds if and only if $\int |f - g| d\mu = \left| \int (f - g) d\mu \right| \Leftrightarrow \int h^+ d\mu + \int h^- d\mu = \left| \int h^+ d\mu - \int h^- d\mu \right| (*)$, where $h = f - g$. Clearly, $(*) \Leftrightarrow \min(\int h^+ d\mu, \int h^- d\mu) = 0$. In other words, at least one of $\int h^+ d\mu$ and $\int h^- d\mu$ is 0. By **Proposition 2.3.9**, $\mu(\{\omega : h^+(\omega) > 0\}) \cdot \mu(\{\omega : h^-(\omega) > 0\}) = 0$, i.e. either $f \geq g, a.e.(\mu)$ or $f \leq g, a.e.(\mu)$.